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LECTURE 4

Tour of Common C++ STL Containers

CS24 · Fall 2026 · Diba Mirza

Standard Template Library (STL) TOUR

ADTs in the STL (also called Containers) store a collection of elements

- | | | |
|---|--------------------------------|--|
| 1 | <code>std::vector</code> | Contiguous, indexable, resizes itself |
| 2 | <code>std::list</code> | Doubly linked, insert or remove anywhere |
| 3 | <code>std::forward_list</code> | Singly linked, one direction only |
| 4 | <code>std::set</code> | Sorted, no duplicates |

C++ TEMPLATES

One function, any type

T is a placeholder -- the compiler fills it in with whatever type you pass.

```
template <class T>
T myMax(T a, T b) {
    return (a > b) ? a : b;
}
```

myMax(3, 7) -- T is int

myMax(3.5, 2.1) -- T is double

myMax('a', 'z') -- T is char

COMPARE THE CONTAINERS

Big-O cheat sheet -- fill it in

Operation	vector	list	forward_list	set
Index [i]				
Insert <i>to back</i>				
Insert <i>to front</i>				
Insert <i>at a position you hold</i>				
Insert <i>a value, sorted + unique</i>				
Find a value				

ANSWER

Big-O cheat sheet

Operation	vector	list	forward_list	set
Index [i]	$O(1)$	n/a	n/a	n/a
Insert <i>to back</i>	$O(1)$ *	$O(1)$	n/a	n/a
Insert <i>to front</i>	$O(n)$	$O(1)$	$O(1)$	n/a
Insert <i>at a position you hold</i>	$O(n)$	$O(1)$	$O(1)$	n/a
Insert <i>a value, sorted + unique</i>	n/a	n/a	n/a	$O(\log n)$
Find a value	$O(n)$	$O(n)$	$O(n)$	$O(\log n)$

* *push_back* is amortized $O(1)$ -- it occasionally resizes and copies everything underneath.